VOUCHSAFE SEQUENTIAL MATCHING HACKATHON

The One Introduction Problem

A supply-constrained sequential matching policy: clarification targeting, outcome-relevant weighting, and variance-aware evaluation

Clarification unlocks the feasible graph. Once it does, nearly every asking rule reaches the structural coverage ceiling—and the remaining score differences are difficult to distinguish from noise.

0.525
INCUMBENT · MSMI PER 100

97–98%
CEILING REACHED BY ASKING RULES

1,326
VALID EPISODES IN THE STUDY

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Starter release 1.0.0
Public synthetic simulator · public seeds
github.com/Yash-Tripath1/sequential-matching-public-simulator-lab

RESEARCH NOTE / NAVIGATION

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RESEARCH INTEGRITY & SCOPE

Declaration. We confirm that this submission represents our team's work and that all external papers, datasets, code, AI tools and other resources used are acknowledged in §8.3 (AI tools and resources) and §8.5 (references). AI tools used: Arena.ai Agent Mode and additional large-language-model chat assistants (ideation, code debugging, drafting and pre-submission review); see §8.3.

Scope. Every person, preference, conversation and outcome in this work is synthetic content from the organisers' public development kit. Nothing here estimates relationship compatibility for real people, and no result below is evidence of product effectiveness. All experiments run against the public simulator on public seeds; no private world, private seed or private organiser file was used. Analysis marked analysis-only reads simulator ground truth and is never imported by, or available to, any policy at inference time.

MSMI = mutual second-Yes, on time, within the date window. All results refer to synthetic worlds from the organisers' public development kit; no result estimates real-world relationship compatibility.

1. Abstract and thesis

We reproduce the incumbent's **0.525** versus **0.350** starter greedy on ten public seeds, then measure how much room is left. Under complete disclosure the reciprocal hard-constraint graph has mean degree **2.53** and **33.9%** of members with no feasible partner; the fraction ever introducible is **0.399** (development), **0.341** (cold start), **0.178** (sparse). Every asking rule reaches **97 to 98%** of that ceiling, so the primary score is supply-determined: asking is a step function (**+0.183**, CI [+0.042, +0.325]), while targeting, weighting and adaptation differences run **−0.000 to +0.117** at a fixed matcher, none conclusive. The pre-specified incumbent versus greedy contrast is **+0.175**, CI [+0.092, +0.258], p = 0.016. An oracle with perfect clarification adds **0.001** coverage and **1.2** assignments: information is not the binding constraint.

At a glance

1. **Scale:** 1,326 valid episodes (1,020 head-to-head, 180 oracle, 108 logging, 18 equivalence); zero invalid actions.
2. **Structure:** coverage ceilings 0.399 / 0.341 / 0.178 and effective-edge caps 118.0 / 90.6 / 23.3; every asking rule at 97 to 98% of ceiling.
3. **What moves:** asking vs none +0.183 [+0.042, +0.325]; incumbent vs greedy (pre-specified) +0.175 [+0.092, +0.258], p = 0.016; weight coding plus global matching +0.133 [+0.008, +0.258].
4. **What does not move:** ask targeting −0.000 to +0.117 (all ns); fitted prior vs hand-coded weights −0.125 [−0.192, −0.067]; online adaptation 0 wins / 60 ties / 0 losses; residual soft asks +0.017 (ns).
5. **Bound:** best screened cell 0.592 is best of 17 and needs a fresh-seed retest; the attainable primary band is about 0.5 to 0.8; a perfect-information oracle adds 0.001 coverage.

Thesis. With reciprocal hard feasibility enforced, MSMI is determined first by whether a policy makes the feasible graph usable at all, a step function that any clarification rule, however naive, saturates; second by how much of the resulting edge supply it consumes before members exit or retire, which we measure at 68 to 87% and which no rule we tested moved; and only third, within a band narrower than 60 episodes can resolve, by how well it ranks the pairs it gets to choose between.

#	Claim	Verdict
H1	Unlock-value ask targeting beats count-based and unranked asking at a fixed matcher	**Not supported**: differences −0.000 to +0.117 at a fixed matcher, all CIs straddling zero
H2	Any asking policy beats no clarification	**Supported**: +0.183 MSMI/100, CI [+0.042, +0.325]
H3	Online adaptation on dense directional responses helps in drift/shift	**Falsified as implemented**: 0/60/0, MSMI outcomes identical to its static version
H4	Waiting/batching helps in supply-rich families and not in sparse	**Untested**, planned (§8.1)
H5	Residual soft-field clarification does not help, because the matching is near-forced	**Not rejected**: +0.017, CI [−0.083, +0.158], p = 0.961
H6	Coverage of any valid policy converges to the structural ceiling	**Supported**: every asking policy at 97 to 98% of it; a perfect-clarification oracle reaches only 0.1 pp more

Contributions. (i) A structural characterisation of the generated worlds (coverage ceilings, edge-supply caps, binding-constraint histogram). (ii) An expected-unlock-value clarification rule with an empirical pass probability prior from the supplied training pools. (iii) A restricted, calibration checked pair-outcome prior fitted on self-logged randomised rollouts, with missing answers treated as unknown rather than disagreement. (iv) A residual-budget soft-field clarification rule no supplied baseline uses. (v) A variance-aware evaluation protocol with seed-block bootstraps, exact sign-flip tests and a power analysis of what the official assessment can resolve.

2. Formalisation

2.1 Notation

An episode runs over decision days $t = 0 \dots 59$ plus 40 follow-up days. One pool holds $|V| = 200$ members; member i has arrival $a_i \le 20$, exit e_i, an always-observable triple $(\text{age}_i, \text{gender}_i, \text{zone}_i)$, and an eleven-field hard bundle H plus seven soft fields S, each with a value, a status in $\{\text{observed}, \text{not_asked}, \text{declined}\}$ and an observation day; $f_{ik} = \varnothing$ when unknown. The primary metric is $\text{MSMI}/100 = 100M/N$ with N the members arrived by day 59 (never-served members stay in N). The official score is the unweighted mean of six family means, each over twenty private seeds. Short display names used in this note map to code identifiers in Appendix A.

2.2 Three-valued reciprocal feasibility

The supplied `eligibility(a, b)` returns one of three values, and the distinction is the hinge of the whole problem:

- **infeasible**: at least one observed hard check fails in either direction. No clarification rescues this pair.
- **needs clarification**: no observed check fails, but at least one hard field is $\varnothing$. The pair is blocked; it may be feasible or infeasible and the policy cannot know which.
- **feasible**: every hard field observed for both members and every reciprocal check passes. Only these pairs may be proposed.

Reciprocity means each check runs twice, once with each member as constraint holder: j's age must lie in i's window **and** i's in j's; zones must be accepted in both directions; one-directional preferences are applied both ways. Two consequences are easy to get wrong. **Declined is absorbing:** a member with one declined hard field can never appear in a feasible pair, on any day, under any policy (**31.9%** of development members, **41.0%** in cold start). **Zones are opaque labels** with set membership only; `relocate` is soft and does not override them.

2.3 The outcome funnel

An introduction produces a chain of five events the spec says must not be collapsed:

$$\text{assigned} \to \text{both respond} \to \text{mutual accept} \to \text{date} \to \text{both second-Yes, on time, in window}$$

MSMI requires a date within 30 days of assignment **and** both Yes answers within 3 days of the date. Our measured stage rates on 7,613 self-logged introductions:

Stage	Estimate	Numerator / denominator	Policy-controllable?
Both respond by deadline	0.5932	4,516 / 7,613	No

Stage	Estimate	Numerator / denominator	Policy-controllable?
Mutual accept, given both responded	0.2573	1,162 / 4,516	**Yes, via pair selection**
Date happens, given mutual accept	0.7685	893 / 1,162	No
MSMI, given a date	0.0907	81 / 893	Partly

The largest single loss is mutual acceptance: 74% of both-responded introductions never reach it (3,354 of 4,516 here; 2,043 of 4,559 losses in the head-to-head funnel, §6.4), and it is the stage pair selection acts on. The two stages outside the policy's control are severe in rate but smaller in count: 41% of introductions never get two responses, and the on-time double-Yes fails for 91% of the dates that reach it (only 204 of 893), the 1 to 5 day answer delay against the 3-day window alone capping that transition at $0.6^2 = 0.36$. **The realistic ceiling on MSMI per assignment is therefore of order** 10^{-2}, which is why a 200-member episode yields about one event and single-episode comparisons are meaningless. MSMI is also self-censoring: a successful pair retires both members, so the pairs that succeed stop being observable.

2.4 What the policy may observe

Only `observe()` output: arrived members with values, statuses and observation days; the introduction list; feedback whose `observed_day` has been reached; the cumulative ask log; the day; remaining budget. No latent preference, no response propensity, no arrival forecast, **no scenario label**, which is why §4.5's non-stationarity problem has to be solved online rather than by looking up the family. The policy additionally receives the public episode seed as a deterministic offset for its own tie-break shuffles (§5.1); the seed is used only to make reruns reproducible, never as a lookup into or inference about outcomes, and no latent or private data crosses the interface. We use exactly this, plus offline-fitted assets derived from our own rollouts on declared training seeds.

3. Structural diagnosis (analysis-only)

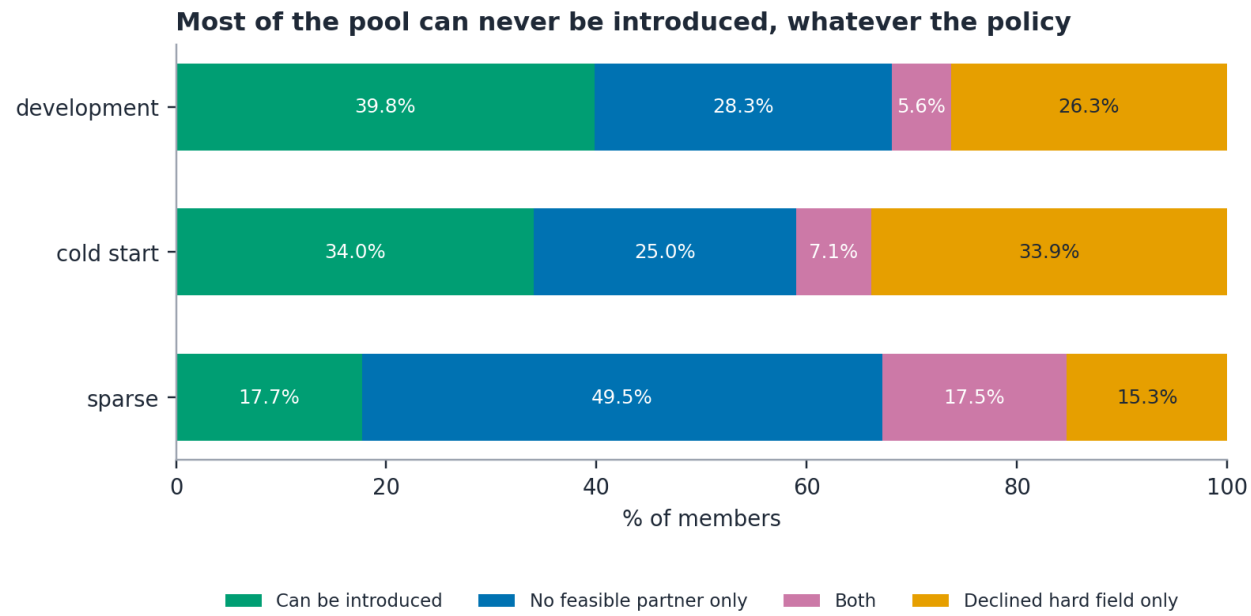
This section reads simulator ground truth to measure the feasible graph. No policy imports it, and none of its numbers is available to a policy at inference time.

The feasible graph is thin. Ten public seeds per family, complete disclosure, members arrived by day 59:

Family	Feasible-pair density	Mean degree	Degree-0 members	Coverage ceiling	Effective edges
development	0.0127	2.53	33.9%	**0.399**	**118.0**
sparse	0.00245	0.49	67.0%	**0.178**	**23.3**
cold start	0.01337	2.66	32.1%	**0.341**	**90.6**
delayed / shift / drift	identical to development	2.53	33.9%	0.399	118.0

Coverage ceiling is the fraction of arrived members with at least one feasible partner who is also never decline-blocked, the value a policy with free, perfect, instantaneous clarification would reach. Effective edges are feasible pairs with neither endpoint decline-blocked, an upper bound on total introductions since pairs cannot repeat within an episode. Full columns (median degree, decline-blocked shares, same-day matchings) and the binding-constraint histogram are in Appendix B.

FIGURE 08



Partition derived from the 7.7 table (overlap counted once).

Figure 8. Most of the pool can never be introduced, whatever the policy: partition of members into introducible, no-feasible-partner, both, and declined-hard-field only. Partition derived from the Appendix B table with the overlap counted once.

In `sparse`, twelve opaque zones with single-zone acceptance put **67.0%** of members at degree zero and leave **23.3** usable pairs in the whole pool; at the measured funnel rates that is about **0.12 MSMI per 100** expected per episode, so the published zeros are what a supply-limited family produces, not a policy failure. In development, `who_to_meet` (74.2%), age window (60.9%) and geography (55.1%) are the three gates that block most pairs (Appendix B), and all three are cheap to reason about because the relevant quantities are always observable.

4. Method

Normalisation, modelling and allocation stay separate, with explicit boundaries so each can be ablated.

4.1 Allocation: maximum-weight matching on a general graph

A member holds **at most one concurrent, outstanding introduction**, so choosing edge (i, j) forecloses every other edge incident on i or j until it resolves: the per-day decision is a matching, and any top-k ranking proposes collisions the simulator must reject. The graph is **not bipartite** (gender preferences need not partition the pool), so we use `networkx.max_weight_matching`, Edmonds' blossom algorithm (§8.5). The per-day objective is $\max_{\mathcal{M}} \sum_{(i,j)\in\mathcal{M}} \hat{p}_{ij}$ with $\hat{p}_{ij}$ estimating $\Pr(\text{MSMI} \mid i, j \text{ assigned})$. **Independence assumption, stated explicitly:** pair outcomes are not exactly independent (the ratio of observed joint acceptance to the product of directional marginals is **1.105**), but given the world the residual dependence is weak enough that expected MSMI is approximately linear in the chosen edges. Two implementation errors we had to fix: **weights must be positive** (a negative score makes the matcher drop an edge whose true probability is small but strictly positive, a pure loss under supply constraint), and **unknown is not disagreement** (coding a known mismatch and an unknown answer both as 0 makes the matcher indifferent between "checked and clash" and "no idea"; our coding is $+1$ / -1 / 0 with the intercept absorbing the unknown case). We do not claim a sequential optimum; waiting is hypothesis H4.

4.2 Clarification: expected unlock value

For each askable member i we score

$$V_i = \frac{1}{3} \sum_{j \in C(i)} \pi_{ij} \cdot \max(0, w_{ij} - u_j)$$

and propose the highest-V members while the daily budget allows. $C(i)$ is the candidate set (same zone, $|\text{age}_i - \text{age}_j| \le 10$, no known hard failure); π_{ij} is the product of the §3.3 marginals over exactly the fields still unknown for this pair, an empirical prior from disclosed answers; w_{ij} is the soft-field matching value (§4.3); u_j is j's opportunity cost, the weight of j's edge in yesterday's matching or 0, making $\max(0, w_{ij} - u_j)$ a reduced profit. Two honest simplifications: true blossom duals are not exposed by `networkx`, so u_j approximates them from the realised matching, and the same-zone / age-gap prefilters discard cross-zone candidates whose unlock probability is $\sim 10^{-3}$ of a same-zone one, an approximation for the 10-second limit that we report as such. **Never re-ask, never impute:** declined hard fields exclude a member from the candidate set; a null is unknown, never zero, never a rejection.

4.3 Modelling: a restricted pair-outcome prior

A randomised logging policy on eighteen declared training seeds (disjoint from the ten evaluation seeds) produced 108 episodes and **7,613 labelled introductions**, 4,516 with both members responding; features are recorded at assignment, labels harvested only after the 40-day follow-up. The target is $\Pr(\text{mutual acceptance} \mid \text{both responded})$ over the 7-day window: a direct MSMI model at a 1.13% base rate achieves held-out AUC 0.581, essentially no discrimination, while the acceptance stage carries the same pair-level signal at 40× the label density. Three nested ridge-logistic specifications were compared on six held-out training seeds; the shipped four-feature **specification B** wins on every held-out measure (log loss **0.5513**, Brier **0.1830**, AUC 0.598), beating both the 7-indicator spec A and the 18-feature spec C, which loses despite four times the parameters because **observability is a confound, not a quality signal**: mutual acceptance is 0.2724 with no soft field known and 0.2317 with all seven known. The shipped coefficients, the full specification search and the held-out calibration table are in Appendix B. **The shipped model is calibrated out of sample** (spec B: held-out log loss 0.5513, Brier 0.1830 on 1,462 pairs from six held-out seeds, central bins within 0.040 of nominal), and §6.2c shows that calibration is not what makes a good edge weight here. **Directional probabilities are not multiplied**: the observed joint rate is 0.2538 against a product of fitted directionals of 0.2297 (ratio 1.105), so we model the joint directly.

4.4 Residual-budget soft clarification

The 12-unit daily budget is **not** binding: measured spend is ≈195 units per episode against a 720 ceiling because 31.9% of development members are decline-blocked, 36.4% arrive complete, and the askable population is exhausted well before the horizon (§6.6). Outcome-relevant soft fields are observed for only ~46% of members in development and ~34% in cold start, and no baseline spends a unit on them. Once hard-bundle asks are exhausted, our rule converts residual units into 1-unit soft answers, prioritising (member, field) pairs by $\deg(i) \times (0.2 + |\beta_k|)$, lifting realised spend from 195 to ~558 units. Whether that converts into MSMI is measured in §6, not assumed.

4.5 Online adaptation on dense proxies

MSMI matures 30+ days after assignment, about one positive label per episode, so an MSMI-only bandit is structurally starved. Directional responses (two per introduction, visible within seven days) give ~120 usable observations by mid-episode; we refit the §4.3 model online by ridge-anchored IRLS towards the offline prior, excluding non-responses from the update, and admit a single `day/60` coefficient, zero in the offline prior, for `drift/shift`, whose measured acceptance falls 0.263 → 0.237 → 0.164 across day blocks. Carried memory (coefficient vector, bounded feature buffer, counters) stays within the 1 MiB cap; we will verify the serialised size before Round 2.

4.6 Audit ledger and deliberate non-actions

Every proposed pair emits the observed evidence per field, unresolved fields, the binding constraint when rejected, the runner-up and its weight, and the reduced profit; explanations keep **model uncertainty**, **missing data** and **stochastic outcome** separate, and no latent value is ever presented as an observation. We deliberately did **not**: read hidden state at inference; build a latent-probability oracle rung (the simulator exposes no latent pair probabilities, so it was dropped); claim inverse-propensity results on the supplied files (propensity is null; doubly-robust evaluation on our own logs is planned); or describe a Thompson-plus-value-of-information hybrid we do not implement.

5. Experimental design

5.1 Worlds, seeds and splits

Use	Seeds	Families	Episodes per method
Prior fitting / model selection	18 declared training seeds: 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79; fitted on 12, evaluated on 6 disjoint held-out seeds (§4.3)	all six	108 logged
Head-to-head evaluation	101, 202, 303, 404, 505, 606, 707, 808, 909, 1010	all six	60

Training and evaluation seeds are disjoint; the split is by world, the only unit that respects the dependence structure, since all 200 members of a pool share one generator draw. Every method sees the identical `(seed, variant)` worlds and policy memory is empty at every episode start. **Randomness is fully declared.** The world generator is seeded by the simulator from `(seed, variant)` and we never touch it. Inside the policy, every draw uses a fixed generator seeded only from public inputs: the shuffled feasible matcher uses `random.Random(17 + day)` and the residual-soft ask tiebreak uses `random.Random(917 + seed)`, where `seed` is the public episode identifier handed to the policy for reproducibility; it enters only as an offset into our own shuffle, is never used to infer an outcome, and the same offsets are used for every configuration so that ties in §6 are ties of rule rather than of draw. There is no wall-clock, no `os.urandom` and no unseeded `random` call in the decision path, so a rerun is bit-identical. Statistical resampling uses `random.Random(20261009)`, fixed independently of the policy. All results are in-process against the public simulator on public seeds, **not** official subprocess or Docker timings.

5.2 Hypotheses and falsification criteria

#	Hypothesis	Falsified if
H1	Unlock-value targeting beats count-based and unranked asking at a fixed matcher	The 95% seed-block CI of the difference excludes 0 in favour of unlock at 10+ seeds per family
H2	Any asking policy beats no clarification on coverage and MSMI	CI includes 0
H3	Dense-response adaptation is no worse anywhere, better in drift/shift	Worse overall with CI excluding 0, or no gain in drift/shift
H4	Daily clearing beats k-day batching in supply-rich families, not in sparse	The ordering is the same in both regimes
H5	Residual soft clarification does not help, because the matching is near-forced	CI excludes 0 in either direction
H6	Coverage of any valid policy converges to the §3 ceiling within 2 pp in development	A policy exceeds the computed ceiling

H5 and H6 are stated as negative predictions on purpose: the structural analysis makes a falsifiable claim about what cannot be improved.

5.3 Statistics and power

The resampling unit is the seed block, one seed's mean across the six families, preserving the within-seed correlation the official score induces. **Primary score** is the unweighted mean of six family means exactly as specified; denominators are never pooled and no best seed is selected. **95% CI** is a percentile bootstrap

over the ten seed blocks (10,000 draws, analysis seed 20261009); **paired tests** are exact two-sided sign-flip permutations (2^{10} arrangements, smallest attainable p = 0.002); **W/T/L** counts over the 60 paired episodes accompany every interval. **No multiplicity correction is applied**; p-values are exploratory and the number of contrasts is stated. The within-method per-episode SD of MSMI/100 is **0.457** (range 0.314 to 0.586), so $n \approx 2\sigma^2(z_{0.975} + z_{0.80})^2/\Delta^2$ gives: 60 episodes per arm resolve about **0.234**; the official 120 episodes about **0.165**, comparable to the entire 0.400 to 0.592 spread we measured; a three-seed pilot about **0.43**, larger than the distance between our best method and the starter greedy, which is why no narrative is built on the pilot's ordering. Episode scores are scaled near-Poisson counts (one event moves the score by 0.5), so the normal formula is an approximation and exact sign-flip p-values plus W/T/L are reported alongside every interval. Validity threats (public seeds only, winner's curse, in-process timing, unadjusted multiplicity, rare events, the label-cutoff bug of §7.5) and their mitigations are tabulated in Appendix D.

6. Results

Scale and replication. 1,020 head-to-head episodes across 17 configurations, 10 public seeds and 6 families, plus 180 oracle-ladder, 108 logging and 18 equivalence episodes: **zero invalid episodes out of 1,326**. Before reporting anything new we checked that our independent harness reproduces the previously published ten-seed screen, exactly to the event count on all five cells (0.350/0.167/0.358/0.408/0.525; 42/20/43/49/63 events), and §6.7 repeats the check decision-for-decision against the organisers' reference `policy.py`.

6.1 The headline is a negative result, and it is the useful one

Every asking configuration saturates the structural coverage ceiling, so the primary score is supply-determined and the differences between sensible asking policies are below what 60 episodes can resolve.

FIGURE 01

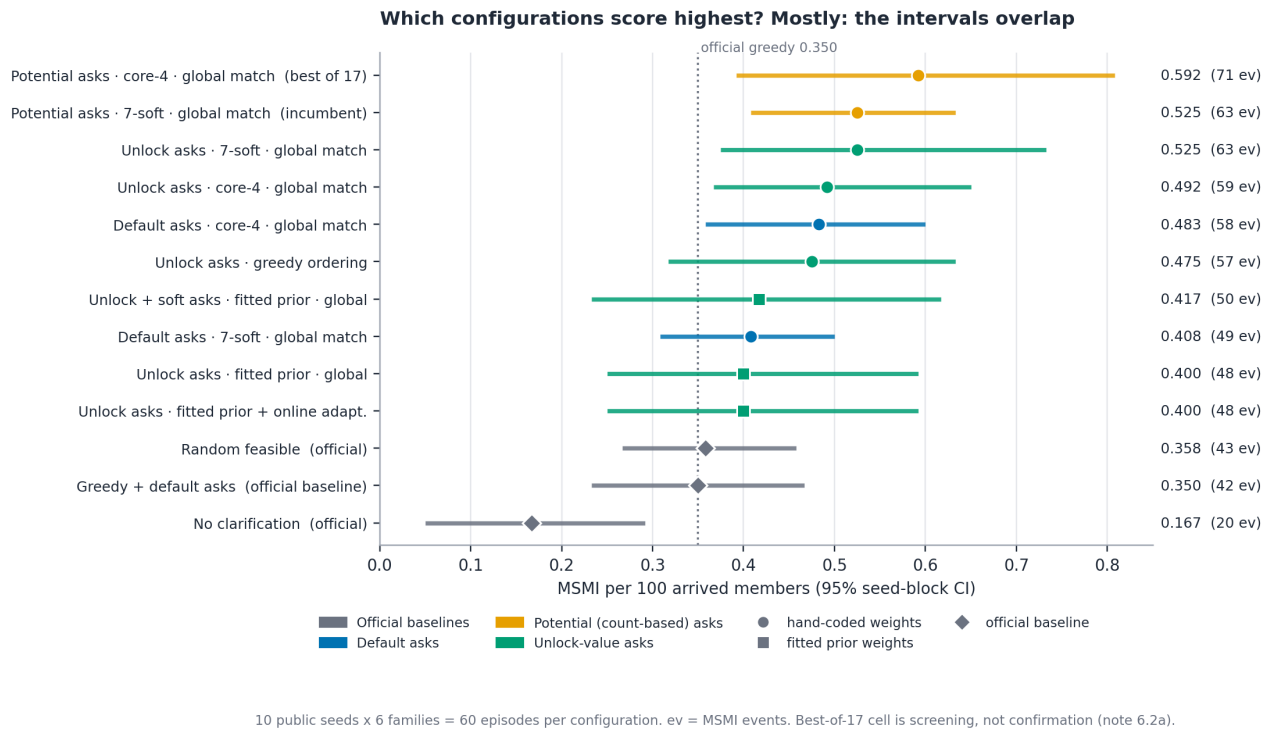
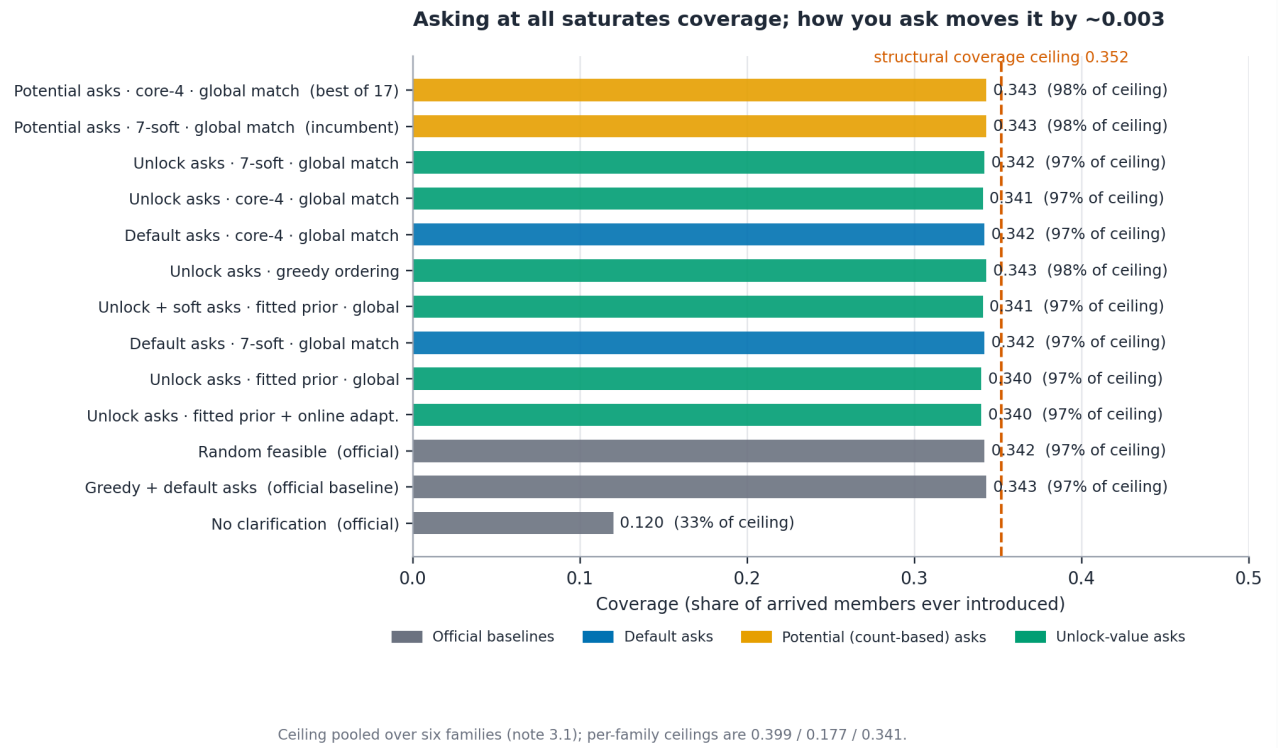


Figure 1. MSMI per 100 arrived members with 95% seed-block CIs for all configurations against the official greedy line. Intervals overlap everywhere except the no-clarification gap; the best-of-17 cell is screening, not confirmation.

FIGURE 02



Ceiling pooled over six families (note 3.1); per-family ceilings are 0.399 / 0.177 / 0.341.

Figure 2. Coverage against the structural ceiling: asking at all saturates coverage (97 to 98% of 0.352 pooled), and how you ask moves it by about 0.003.

Read down the assignments column of the full table (Appendix B): across the sixteen asking configurations, total introductions over 60 episodes move from 4,555 to 4,681 and coverage from 0.3399 to 0.3432, and every one of them sits at **97 to 98% of the §3 ceiling**, including the unmodified starter greedy. That confirms H6 and changes what the problem is: clarification is a **step function**, not a lever skilled targeting pulls further. Asking at all takes coverage 0.120 to ~0.342 and MSMI/100 0.167 to ~0.45; asking cleverly takes it nowhere measurable, because the ceiling is a property of the generated world. Only `no_ask` sits away from the ceiling at 33% of it; its contrast against the starter greedy is **+0.183 MSMI/100, 95% seed-block CI [+0.042, +0.325], sign-flip p = 0.055, 22 wins / 33 ties / 5 losses**, 42 events against 20. H2 is supported: the bootstrap CI excludes zero, and the support is structural rather than a single p-value, since **all sixteen asking-versus-no-ask contrasts among the 136 enumerated pairwise contrasts have intervals excluding zero**, from +0.183 to +0.425 (full enumeration in `results/analysis.json`).

Fresh-seed split. Excluding the three pilot seeds on which the incumbent was first chosen, incumbent versus official greedy on seeds 404 to 1010 alone is **+0.143 MSMI/100, 95% seed-block CI [−0.009, +0.295], 4 wins / 3 ties / 0 losses** over the seven seeds, computed from `results/scaled_factorial_10seeds.json`. The direction survives off the pilot seeds; at seven seeds the interval still straddles zero, so this remains a screen, and the ten-seed pre-specified +0.175 remains a recomputation on selection data.

Soft-ask screen, pilot to holdout. The residual soft-ask rule was screened on the pilot and then tested held-out: on the three pilot seeds it added **3 MSMI events** (selection data, descriptive only); on the seven held-out seeds 404 to 1010 it gives **+0.024 MSMI/100, 95% seed-block bootstrap CI [−0.071, +0.131], exact sign-flip p = 0.844, 42 vs 40 events**, at about **+83 ask units** per episode with virtually unchanged coverage. This is the cleanest pilot-to-holdout result in the note: a small, statistically null gain bought with a third more asking.

The negative result is made precise by the same enumeration: among the thirteen configurations that both ask and match, only three of 78 pairwise intervals exclude zero, all the same comparison in different cells, **a fitted-prior edge weight losing to a hand-coded one** (§6.2c). Everything else is pair quality on a fixed volume, bounded twice: by a mean-degree-2.5 matching with little room to choose, and by a funnel whose largest loss, mutual acceptance (45% of the 4,559 non-MSMI assignments), is where pair selection acts, while the policy-uncontrollable stages (non-response 41%, on-time answers 9%) set the floor.

6.2 The factorial: what moved and what did not

Asking and the weight coding move the score; ask targeting, fitted probabilities and online adaptation do not.

FIGURE 06



Figure 6. All twelve pre-specified contrasts with 95% seed-block CIs: only asking-vs-none, the pre-specified incumbent contrast and the best-of-17 cell exclude zero on the positive side, and the fitted-prior-vs-core-4 contrast in the potential-ask cell excludes it on the negative side.

(a) The best cell is potential asks + signed core-4 at 0.592 (71 events), +0.242 versus official greedy, CI [+0.125, +0.383], p = 0.008, 27/27/6. We state the selection problem first: it is the best of seventeen screened configurations, p = 0.008 clears neither the 0.05/12 ≈ 0.004 nor the 0.05/136 ≈ 0.0004 Bonferroni threshold, and 60 episodes resolve about 0.234 against an effect of 0.242. The defensible claim is that this configuration scored 0.592 on ten public seeds and must be re-tested on fresh seeds before it is believed.

(b) The matcher improvement decomposes into two halves, neither significant alone. Global matching vs greedy ordering at fixed weights: +0.058, CI [−0.058, +0.167], p = 0.445. Signed core-4 vs 7-soft agreement count: +0.075, CI [−0.025, +0.192], p = 0.328. Together +0.133, CI [+0.008, +0.258], p = 0.102, equal to the sum of the halves to three decimals. The coding half is not a model: it fixes treating a known disagreement as no information, and letting weights go non-positive so the matcher discards feasible edges.

(c) Ask targeting does not help, and the fitted prior actively hurts. At a fixed matcher, unlock-value vs count-based potential asking is −0.000 (CI [−0.167, +0.200], 17/26/17, both 63 events); vs default asking it is +0.117 at 7-soft weights but +0.008 at core-4 and −0.000 at the fitted prior. **H1 is not supported.** The fitted logistic prior is below the hand-coded signed rule in all three ask cells, and in the potential-ask cell the gap excludes zero: **−0.125, CI [−0.192, −0.067], p = 0.008, 2/43/15**. The direction is consistent across all three ask rules, which is harder to dismiss than one cell.

Why a calibrated probability loses. A calibrated $\hat{p}$ is the theoretically correct single-day edge weight, but `max_weight_matching` with `maxcardinality=False` is scale-sensitive: it trades matching more pairs against matching better pairs, and that trade depends on the spread of the weights. Our prior maps every feasible pair into $\hat{p} \in [0.23, 0.56]$, a band so narrow the optimiser behaves almost like maximum-cardinality matching; the signed core-4 rule spans $[0.06, 0.98]$ and the 7-soft count $[0.01, 7.01]$, imposing a real quality/cardinality trade. That a correctly calibrated probability (spec B, held-out log loss 0.5513, Brier 0.1830, central calibration bins within 0.040 of nominal; Appendix B) underperforms a mis-scaled heuristic is **consistent with a weight-scale mechanism**; the separable test (rerun the prior under a monotone spread or rank transform) is named Round 2 work (§8.1), as is the deeper argument that the myopic single-day objective is misspecified under eight-day busy windows, non-repeatable pairs and permanent retirement.

6.3 Scenario-level results: one family reverses the ordering

Sparse is supply-limited for everyone, and cold start is the one family where untargeted asking beats our targeted rule.

FIGURE 03

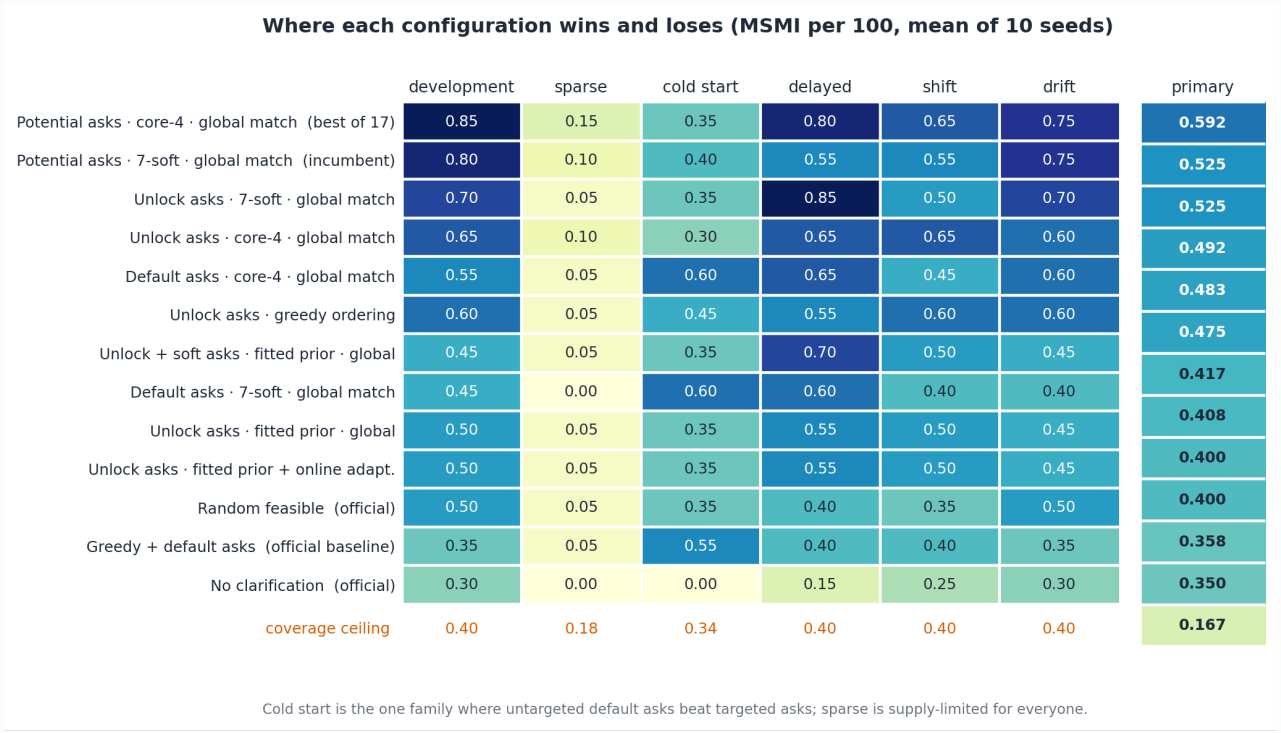


Figure 3. Family-level heatmap: every configuration collapses in sparse, the official greedy wins cold start, and the primary column spans 0.167 to 0.592 against per-family ceilings of 0.40 / 0.18 / 0.34 / 0.40 / 0.40 / 0.40.

Sparse scores 0.00 to 0.15 for every configuration, consistent with the ~0.12 expected from 23.3 usable pairs and the measured funnel; a zero is not an equivalence result. Cold start reverses the development ordering: default-ask cells reach 0.600 where unlock-value cells reach 0.300 to 0.350 (about 12 vs 6 events

in this one family; exploratory), explained in §7.3. Shift and drift do not reward the offline prior, exactly as §4.5 predicted, and the online correction did not repair this (§6.5). The full per-family table is in Appendix B.

6.4 The funnel: volume is identical, conversion differs in the noisiest stage

All sixteen asking configurations make the same introductions; only the on-time double-Yes stage separates them, and its spread is of the order of sampling noise.

FIGURE 04

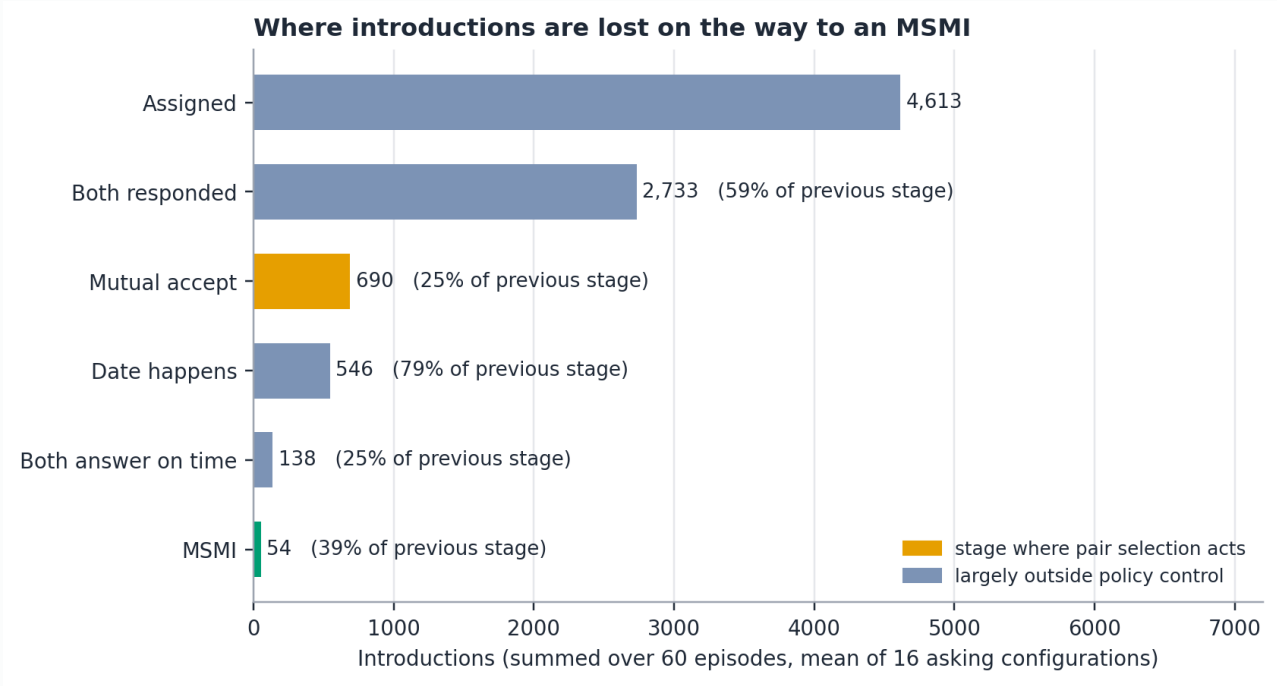


Figure 4. Where introductions are lost on the way to an MSMI, summed over 60 episodes and averaged over the 16 asking configurations: the mutual-acceptance stage is the largest loss and the one pair selection acts on.

P(both respond) spans 0.585 to 0.617 across methods and P(date | mutual) 0.75 to 0.80, matching the 0.769 measured on 1,162 logging mutual acceptances: both are constants of the world. P(mutual | responded) spans a 2.2-point band; P(both on time | date) spans 0.228 to 0.305 on 515 to 577 dates, where the binomial SD of one such rate is 0.019 and the expected range of thirteen draws about 0.06, close to the 0.077 observed. The stage that separates top from bottom is therefore the one least connected to anything a matcher controls, and we do not interpret the 71-vs-42 event gap as proof that any weighting scheme is better; it is a candidate that survived one screen. The loss structure is unambiguous: of the **4,559** non-MSMI assignments, **1,880 (41%)** are lost at response, **2,043 (45%)** at mutual acceptance, **144 (3%)** at the date stage, **408 (9%)** at the on-time answer stage and **84 (2%)** at the final both-Yes step. Per-method funnel counts and stage rates are in Appendix B.

6.5 Oracle ladder: perfect information buys almost nothing

An oracle with free, complete, instantaneous disclosure of every non-declined field adds 0.001 coverage and 1.2 assignments over uniform random feasible pairing: information is not the binding constraint.

FIGURE 05

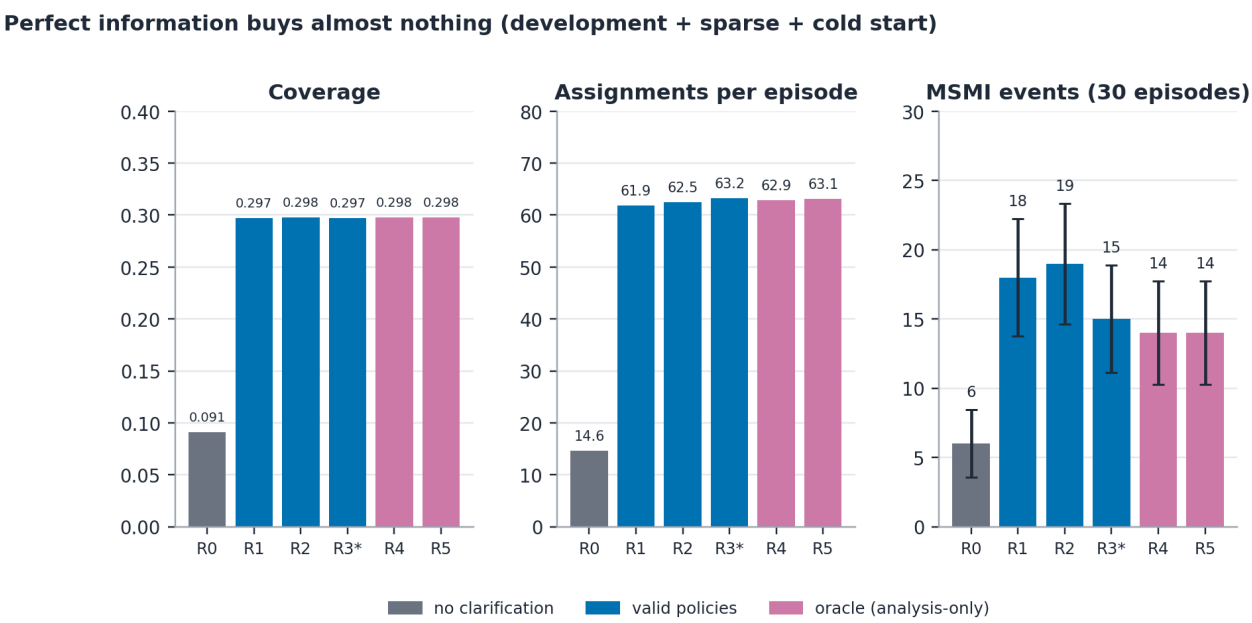


Figure 5. Oracle ladder on 30 dev/sparse/cold worlds per rung: coverage and assignments are flat from R1 to R5 while mutual-acceptance quality rises about 12% and does not convert; whiskers are Poisson SDs on the 30-episode event counts.

From R1 (uniform random feasible) to R5 (oracle plus maximum-cardinality matching), coverage moves 0.297 to 0.298 and assignments 61.9 to 63.1, on a ladder spanning 14 to 19 events per 30 episodes where the Poisson SD at 16 events is 4; only R0 (no clarification, 6 events) is separable. Perfect information raises P(mutual | responded) from 0.245 to about 0.275, a 12% relative gain on ~1,100 both-responded pairs per rung, **and it does not convert**: MSMI per assignment falls 0.0097 to 0.0074 because downstream stages move the other way within their noise. Two caveats, because this table is the one most easily misread. R3 carries the fitted prior (the weakest weight rule, §6.2c), so R3 scoring below R2 reproduces §6.2c inside the ladder rather than indicting the method; and the ladder covers three families, so its means are internal comparisons on identical worlds, not six-family primaries. Two conclusions survive both caveats: no clarification strategy closes a gap the oracle does not close, which is why the §6.2 targeting nulls should have been expected; and a surrogate (mutual acceptance, the first tie-break) can move while the primary does not, so reporting the surrogate alone would have produced a confident false claim. Rung values and the per-family ladder are in Appendix B.

6.6 Cost, saturation and reproducibility

The ask budget is not binding and never was: the askable population saturates by day 15 to 20, leaving 40 days of budget with nobody left to ask. Every hard-only rule spends about 195 of 720 units (soft-ask rules ~582 to 614), and the saturation probe (Appendix B, Figure 7) shows every ranked rule reaching the same 140-member plateau in development (132 in sparse) by day 15 to 20, while `no_ask` plateaus at 75: clarification moves the usable graph from 75 to 140 members, exactly the H2 step, and live feasible edges collapse to single digits by day 10 under every rule. The binding constraint is that the feasible graph is consumed faster than it is replenished, which is why §8.1 prioritises waiting and batching, the only lever acting on the timing of consumption. Cost and waiting-time accounting (mean first-introduction wait 6.3 days for hard-only rules vs 3.95 for `no_ask`, whose short wait is bought with 12% coverage; unserved members 131.6 vs 176.0 of 200) is in Appendix B. **Reproducibility of the supplied baselines:**

our in-process harness reproduces the organisers' reference `policy.py` decision-for-decision on seeds 101 to 606 across all six families: every ask list, pair batch and final metric identical for `greedy`, `no_asks` and `random`.

7. Failure analysis

7.1 Sparse geography: supply, not policy

Twelve opaque zones with single-zone acceptance put 67.0% of sparse members at degree zero and leave 23.3 usable pairs; consuming every one of them at the measured funnel yields **0.122 MSMI per 100** against **0.619** for a development world at full utilisation, so sparse is a floor on the primary score, not a test of skill. Combining the edge-supply caps with the funnel gives an arithmetic **bound on the primary score** at different pair-selection qualities (Appendix B): 0.512 at our measured logging quality, 0.606 at the incumbent's published quality, up to 1.009 at best-case selection. This is a bound, not a forecast: it assumes every effective edge is consumed and ignores timing. Our randomised logging policy consumed 71% of the edge supply in development (87% in sparse) and scored 0.375 against the 0.512 bound, so the honest reading is that the achievable primary sits between roughly **0.5 and 0.8**, and both the published incumbent (0.525) and our best screened cell (0.592) are inside that band. Any claim of a large multiple over the incumbent should be treated with suspicion until the bound is beaten, not merely approached. The correct sparse strategy is selective, not expansive: with 23 pairs, revealing the good ones early beats revealing all of them, because members exit. Caveat: only three distinct sparse worlds exist among the ten public seeds.

7.2 Withheld answers: a permanently excluded third of the pool

Because declined is absorbing, **31.9%** of development and **41.0%** of cold-start members can never appear in a feasible pair; unioned with degree-zero isolation the never-introducible share is 60.2% / 66.0% / 82.3% across development / cold start / sparse (Figure 8, Appendix B). Three consequences: coverage must be reported against the ceiling, not against 1.0 (our best configuration reaches 0.465 on development seed 101, equal to that world's per-seed ceiling); clarification spent on decline-blocked members is destroyed, and every ask rule we test filters them first; and missingness here is informative about availability, not preference (acceptance is lower when more soft fields are observed, 0.2724 to 0.2317), so any model encoding observability as a feature learns a spurious preference; the fix is to exclude the indicator, not re-weight by it.

7.3 Cold start: where our own ask rule loses

Cold start pre-observes the hard bundle for only 20.6% of members (36.4% elsewhere), has the highest decline-blocked share (41.0%), and ~34% soft-field observation, so the weights w_{ij} that drive V_i are computed from mostly-empty inputs and the ranking degenerates to $\sum_j \pi_{ij}$ with more noise. This is the sharpest falsification of H1: the advantage of a value-ranked ask rule depends on having enough observed soft signal to rank with. The named fix is a **shrinkage rule** (blend V_i towards the count-based score as observed soft fraction falls), listed as planned work with its test: does the cold-start reversal disappear under shrinkage?

7.4 Delayed feedback and the follow-up window

In `delayed`, date offsets reach 33 days against the 30-day window, so some introductions with a date and two timely Yes answers still fail the outcome definition; our label helper checks `date_day − assigned_day ≤ 30` explicitly and counts the two failure routes separately. Right-censoring is handled by construction: la-

bels are harvested after the 40-day follow-up and unresolved introductions return `None`, excluded from learning rather than coded 0. The operational consequence: **feedback arrives too late to help the episode that earned it**; a day-40 introduction matures after day 70, past the horizon, which is the whole argument for §4.5.

7.5 An internal audit: the label-cutoff bug

Our earlier helper labelled a date plus two timely Yes answers as MSMI positive **without checking the 30-day window**; one audited instance (delayed, seed 303: assigned day 31, date day 62). The simulator's own scoring was always correct, so no reported score was affected; only online-learning updates could be corrupted. Consequences we accept: every learning-based row from the three-seed pilot (Thompson 0.472, GP-UCB 0.389, GP-mean 0.222) is provisional and not cited as a result; the corrected label is used for all of §4.3 and §6; we replaced the old learners rather than re-running them.

7.6 Competition for the same candidate

At mean degree 2.53 competition is the norm; a max-weight matching resolves it globally while the starter greedy resolves it by sort order. Two dynamic forms remain named, not solved: **temporal competition** (introducing i to j today consumes the pair permanently and makes both busy for eight days; if i has a better partner currently busy or not yet arrived, waiting dominates; H4) and **success consumes supply** (a mutual second-Yes retires both members, so maximising MSMI spends the scarcest resource in the system, and the optimal policy may be more selective than the single-day objective implies; quantifying this needs a two-stage look-ahead we have not implemented).

8. Round 2 plan, disclosure and references

8.1 What is done, what is planned

Item	Status
Structural diagnosis; pass probability prior; 108-episode randomised log; restricted pair-outcome prior; unlock-value rule; residual-soft rule	**Done**
Online ridge-anchored adaptation on directional responses	**Done and falsified**: MSMI outcomes identical to its own static version in all 60 episodes while six assignments differed (4,658 vs 4,652; §6.5); kept as an instrumented null
Head-to-head §6, funnel, oracle ladder, baseline equivalence	**Done**
Cold-start shrinkage rule; waiting/batching ablation $k \in \{2, 3, 5\}$; exact blossom duals; doubly-robust OPE on own logs; two-stage look-ahead; JSON protocol adapter and container timing; 20-seed public runs and 1 MiB check	**Planned**
Fitted prior under a monotone spread/rank transform (scale vs misspecification, §6.2c)	**Planned**
Latent-probability oracle rung	**Dropped** (§4.6)

8.2 Round 2 build order

1. Wrap the winning configuration in the JSON protocol; handle empty population and empty feasible graph explicitly.
2. Ship `hard_priors.json` and `outcome_prior.json` as declared inference assets; confirm the 2 GiB image and 1 MiB memory caps.
3. Containerise and measure; the $O(n^2)$ eligibility pass (~20k pair checks per phase at 200 members) is the term to watch against the 10-second limit.
4. Run the cold-start shrinkage rule and the batching ablation; keep whichever survives a paired test at 20+ seeds.
5. Publish per-episode machine-readable results for the four required comparisons on identical seeds, plus funnel and waiting-time tables, pinned with SHA-256.

8.3 AI tools disclosure

This note and the accompanying code were produced with the assistance of large-language-model coding agents, used for: reading and summarising the starter kit and its documentation; drafting and debugging the experiment harness, the policy implementations and the analysis scripts; and drafting and critically revising this document. Separate chat-assistant sessions (Claude- and Gemini-family models, via Arena.ai Agent Mode and other tools) additionally contributed ideation, cross-checking of the draft against its own tables, and pre-submission review; every such suggestion was accepted only after arithmetic or artefact cross-check, and the final decisions rest with the team. All experimental numbers reported here were produced by executing the committed Python code against the organisers' public simulator release 1.0.0 on public seeds, and every table is reproducible from the scripts named in the README's reproduction section. No model was used to generate results, and no result was accepted without a corresponding artefact on disk. One model-assisted implementation contained the label-cutoff defect of §7.5; it was found by human-directed audit of the code against §8 of the problem statement, not by the model. Where an earlier AI-

generated draft contained claims contradicted by our own audit (bipartite/Hungarian allocation, zone adjacency, "one introduction per person per day", regret curves, exact value of information, a Thompson-plus-VOI hybrid), those claims are removed here and asserted nowhere in this note.

8.4 Synthetic-data disclaimer

All members, profiles, preferences, conversations, introductions and outcomes are newly generated or authored synthetic content; demographic proportions and outcome behaviour are invented assumptions, not fitted distributions of real people. Simulator performance is an engineering and research test, not evidence that any model can predict a relationship, and nothing here should be presented as product effectiveness. Specifically, these invented outcomes cannot establish that any feature we weighted predicts real compatibility (we recovered the generator's decision, not the world's); that a higher MSMI is a better product outcome (MSMI is a synthetic proxy whose funnel is dominated by invented response and punctuality rates); that clarification is acceptable to real people (a simulated member never experiences being asked, declining, or objecting to timing); or anything about fairness (the generator draws acceptance uniformly and independently of demographics, so our runs contain no bias to detect, and absence of measured disparity here is not evidence of absence in production). Real use would require, in order: a schema adapter with an audited missing-answer rule; a consent layer with per-question withdrawal, since declined must be respectful in production as it is absorbing here; verified-attribute eligibility with a human review path; outcome labels from actual consented follow-up, which would face the censoring problem of §2.3; prospective validation powered on the event rate that actually occurs; and a fairness audit across protected attributes, which this simulator structurally cannot supply. None of that is started. The method is portable (three-valued feasibility algebra, supply diagnosis, matching formulation, seed-block statistics), but every number in §6 is a property of a generator we did not write.

8.5 References

All works below were located and checked against their publishers' records on 9 to 10 October 2026; volume and page numbers are as printed there (Edmonds carries no issue number in the publisher record). We have not fabricated a citation to fill a gap, and we cite nothing we have not located.

- **Akbarpour, M., Li, S., & Oveis Gharan, S. (2020).** "Thickness and Information in Dynamic Matching Markets." Journal of Political Economy, 128(3), 783 to 815. DOI 10.1086/704761. The Patient algorithm that waits to thicken the market loses exponentially less than Greedy, but **only when the planner can identify agents about to depart**; when departure times are unobservable, Greedy is close to optimal. That precondition is precisely the empirical question our §6.6 saturation probe answers here: feasible edges vanish because members are consumed, retire permanently, or age out at day 60, and the first two are visible in the observation contract. H4 is therefore the one lever the theory says could still be large, and §8.2 schedules it.
- **Edmonds, J. (1965).** "Paths, Trees, and Flowers." Canadian Journal of Mathematics, 17, 449 to 467. DOI 10.4153/CJM-1965-045-4. The blossom algorithm, the correct general-graph maximum-matching primitive for a pool that is not bipartite, and the reason §4.1 uses `networkx.max_weight_matching` rather than a Hungarian assignment.
- **Dudík, M., Langford, J., & Li, L. (2011).** "Doubly Robust Policy Evaluation and Learning." Proceedings of the 28th International Conference on Machine Learning (ICML '11), 1097 to 1104. The estimator we plan to apply to our own randomised logging rollouts, where propensities are known by construction (§8.1).
- **Russo, D., Van Roy, B., Kazerouni, A., Osband, I., & Wen, Z. (2018).** "A Tutorial on Thompson Sampling." Foundations and Trends in Machine Learning, 11(1), 1 to 96. Background for the label-starvation argument in §4.5: at about 0.9 MSMI events per episode, a posterior over pair outcomes cannot be updated often enough to matter inside a 60-day episode.

- **Hitsch, G. J., Hortaçsu, A., & Ariely, D. (2010).** "Matching and Sorting in Online Dating." American Economic Review, 100(1), 130 to 163. DOI 10.1257/aer.100.1.130. Motivation for treating introductions as a two-sided allocation problem with competition for the same candidates (§7.6).
- **Karp, R. M., Vazirani, U. V., & Vazirani, V. V. (1990).** "An Optimal Algorithm for On-line Bipartite Matching." Proceedings of STOC '90, 352 to 358. DOI 10.1145/100216.100262. Foundational online-matching background; its bipartite one-sided-arrival model does not transfer to this simulator's general-graph, reciprocal-feasibility setting, and we do not imply that it does.
- **Gamlath, B., Kapralov, M., Maggiori, A., Svensson, O., & Wajc, D. (2019).** "Online Matching with General Arrivals." Proceedings of FOCS 2019, 26 to 37. DOI 10.1109/FOCS.2019.00011. The closest model-family reference for general arrivals; our daily batch decisions and clarification actions still make the setting distinct.
- **Saar-Tsechansky, M., Melville, P., & Provost, F. (2009).** "Active Feature-Value Acquisition." Management Science, 55(4), 664 to 684. DOI 10.1287/mnsc.1080.0952. Cost-aware information-acquisition context for the clarification problem; an analogy for budgeted questioning, not a direct matching algorithm.
- **Vouchsafe / Romeo & Juliet (2026).** The One Introduction Problem, final participant specification 1.0.0, with the bundled `docs/DATA_CONTRACT.md`, `docs/POLICY_INTERFACE.md` and `docs/SUBMISSION.md`. Authoritative for every rule quoted in this note; where an earlier draft of our own work conflicted with it, the specification won and the draft was corrected (§8.3).

8.6 Data and software provenance

- Simulator and data: organisers' public synthetic release 1.0.0; the bundled `starter/kit.py` is unchanged and records upstream commit `a8e26b35118cfa8e886a02f93984923e43ab64f6` (`starter/SOURCE.md`, `starter/LICENSE`, `starter/DATA_LICENSE.md`). Every member and outcome is synthetic (§8.4). No private seeds, worlds, organiser files or real-user data were used; the only external data are the six supplied training pools used for the §3 marginals.
- Experiment code carries no separate top-level license; the scripts are those named in the README reproduction section. The three official baselines were re-verified through the organisers' reference evaluator in trusted-local subprocess mode (54/54 episodes identical; `research/VERIFICATION_2026_10_09.md`).
- The three-seed soft-question screen (72 episodes) and the seven-seed follow-up (42 episodes) were run on the author's Ubuntu VM. The shared bundle contains the aggregate soft-question results but not the VM-generated row-level JSON; this is documented in `research/PROVENANCE_AND_ARTIFACT_STATUS.md`.

8.7 Computational environment and reproducibility

All core factorial, head-to-head and oracle-ladder episodes ran in-process; the exact execution host of those runs is not recorded in this bundle and no hardware specification is claimed. Wall times in Appendix B are in-process seconds per episode, not official per-invocation or Docker timings; container validation is Round 2 work (§8.2). Dependencies are pinned in `requirements.txt`; the starter kit is standard library only. Reproducibility anchors: every policy random draw is seeded from public inputs only (§5.1); bootstrap intervals use analysis seed 20261009; reruns are bit-identical by construction. The full script inventory and output map (§9 of earlier drafts) now live in `README.md`.

Appendix A. Code-name key

Short name used in this note	Code identifier
Potential asks · core-4 · global match (best of 17)	`lab_potential_core4_maxw`
Potential asks · 7-soft · global match (incumbent)	`lab_potential_maxw`
Unlock asks · 7-soft · global match	`unlock_soft7_maxw`
Unlock asks · core-4 · global match	`unlock_core4_maxw`
Default asks · core-4 · global match	`defaultask_core4_maxw`
Unlock asks · greedy ordering	`unlock_greedy`
Unlock + soft asks · fitted prior · global	`unlocksoft_prior_maxw`
Default asks · 7-soft · global match	`defaultask_soft7_maxw`
Unlock asks · fitted prior · global	`unlock_prior_maxw`
Unlock asks · fitted prior + online adapt.	`unlock_prior_online_day`
Unranked full budget · core-4 · global	`fullbudget_unranked_maxw`
Random feasible (official)	`random_feasible`
Greedy + default asks (official baseline)	`greedy`
No clarification (official)	`no_ask`

Appendix B. Full tables

B.1 Structure under complete disclosure

Ten public seeds per family, complete disclosure, members arrived by day 59:

Family	Feasible-pair density	Mean degree	Median degree	Members with degree 0	Decline-blocked	Coverage ceiling	Effective edges	Max same-day matching
development	0.0127	2.53	2	33.9%	31.9%	**0.399**	**118.0**	34.4
sparse	0.00245	0.49	0	67.0%	32.8%	**0.178**	**23.3**	15.3
cold start	0.01337	2.66	1	32.1%	41.0%	**0.341**	**90.6**	28.6
delayed / shift / drift	identical to development					0.399	118.0	34.4

B.2 Binding constraints (development, seed 101, all 19,900 pairs)

Blocking reason	Pairs blocked	Share of 19,900
`who_to_meet`	14,757	74.2%
age window	12,116	60.9%
geography	10,974	55.1%
`wants_children` conflict	4,761	23.9%
smoking	4,545	22.8%
`relationship_structure`	3,600	18.1%
`partner_children`	2,970	14.9%
`schedule`	2,684	13.5%
feasible	**273**	**1.37%**

Reasons overlap, so shares do not sum to one.

B.3 Empirical pass probability prior (six training pools only)

Unknown field	Pass probability	Notes
`who_to_meet`	0.494 / 0.480 / 0.461	by partner gender: woman / man / non-binary
`acceptable_zones`, same zone	**1.000**	own zone is always accepted
`acceptable_zones`, different zone	**0.0756**	only the 22.9% of members with two zones can pass
`schedule` overlap	0.856	
`relationship_structure` equal	0.793	88.3% monogamous
`wants_children` no conflict	0.755	yes 0.362 / no 0.339 / unsure 0.300
smoking acceptable	0.878	50.4% accept any; 75.4% are non-smokers
`partner_children` acceptable	0.921	15.5% have children
age window	~0.13	marginal used by the unlock prior, see note

Age-window conditioning. Age is always observable, so the age window is never an unknown field in the eligibility check; §B.2's figure (age blocks 60.9% of fully-disclosed pairs, a ~0.39 pass rate over all 19,900 pairs) is measured on the full pair population, while the ~0.13 row is the marginal the unlock-value rule uses when scoring blocked candidate pairs under the §4.2 prefilter (same zone, |age gap| ≤ 10), estimated from the six training pools. The two numbers answer different questions and must not be compared directly. The same-zone / different-zone asymmetry is the sharpest fact in the table: a blocked cross-zone pair carries roughly $0.076^2 \approx 0.006$ of the unlock probability of a blocked same-zone pair.

B.4 Pair-outcome prior: specification search, coefficients, calibration

Specification	Features	Held-out log loss	Null	Brier	AUC
A	7 agreement indicators	0.55198	0.56650	0.18337	0.595
B (shipped)	**4 core agreement indicators**	**0.55125**	0.56650	**0.18298**	**0.598**
C	A + 7 both-known + age gap + same-zone + day	0.55722	0.56650	0.18561	0.574

Specification C loses on every held-out measure despite four times the parameters, because missingness is a confound: mutual acceptance is **0.2724** with no soft field known, 0.2546 with one to three known, **0.2317** with all seven known; the seven both-known indicators are near-collinear and receive large offsetting coefficients (one reached +0.61) that encode who is observable, not who is compatible; and `same_zone` has 4,492 observations on one side and **24** on the other among logged pairs, so its coefficient is unidentifiable.

Coefficient	Value
intercept	−1.0065
`relationship_pace`: agree	+0.5328
`relationship_goal`: agree	+0.4212
`conversations`: agree	+0.2958
`lifestyle`: agree	−0.0702

Calibration on held-out data (target: mutual acceptance given both responded within 7 days; fitted on 12 seeds, 3,054 both-responded pairs; evaluated on six disjoint held-out seeds, 1,462 pairs, base rate 0.2538). The table reports the extended specification C; the shipped specification B has held-out log loss **0.5513** and Brier **0.1830** on the same split:

Predicted bin	n	Mean predicted	Observed rate	Gap
0.05 to 0.10	70	0.0897	0.1571	**+0.067**
0.10 to 0.20	248	0.1694	0.1532	−0.016
0.20 to 0.30	784	0.2631	0.2640	+0.001
0.30 to 0.40	308	0.3319	0.2922	−0.040
0.40 to 0.60	51	0.4736	0.4706	−0.003
0.60 to 0.80	1	0.6003	1.0000	(n = 1, uninformative)

The largest absolute gap in a bin with n ≥ 50 other than the lowest is 0.040, and the three central bins holding 1,340 pairs are within 0.040 of nominal; the one real defect is the lowest bin, where ridge over-shrinks strongly disagreeing pairs toward zero. Held-out log loss 0.5572, Brier 0.1856 for the extended specification shown. The fitted prior is calibrated, and calibration is not what makes a good edge weight here (§6.2c).

B.5 Power table

Gap to detect (MSMI/100)	Episodes per arm	Equivalent seeds × 6 families
0.50	14	2.3
0.30	37	6.2
0.20	83	13.8
0.15	146	24.3
0.10	329	54.8

B.6 Head-to-head configuration table

60 episodes per configuration, ten seeds × six families. Ceiling is the §3 structural maximum on the same worlds; % of ceiling is coverage divided by it, pooled across families as a summary figure (per-family ceilings differ, §3); `results/primary_summary.csv` carries the per-seed coverage-to-ceiling ratios once added on the working machine.

Method	MSMI/100	95% seed-block CI	Events	Coverage	Ceiling	% of ceiling	Assignments	Ask cost
`lab_potential_core4_maxw`	**0.592**	[0.392, 0.808]	71	0.343	0.352	98%	77.1	197
`lab_potential_maxw`	**0.525**	[0.408, 0.633]	63	0.343	0.352	98%	76.4	197
`unlock_soft7_maxw`	**0.525**	[0.375, 0.733]	63	0.342	0.352	97%	76.1	195
`unlock_core4_maxw`	**0.492**	[0.367, 0.650]	59	0.341	0.352	97%	77.2	195
`defaultask_core4_maxw`	**0.483**	[0.358, 0.600]	58	0.342	0.352	97%	76.9	197
`unlock_greedy`	**0.475**	[0.317, 0.633]	57	0.343	0.352	98%	76.2	195
`unlocksoft_prior_maxw`	**0.417**	[0.233, 0.617]	50	0.341	0.352	97%	77.8	582
`defaultask_soft7_maxw`	**0.408**	[0.308, 0.500]	49	0.342	0.352	97%	76.6	197
`unlock_prior_maxw`	**0.400**	[0.250, 0.592]	48	0.340	0.352	97%	77.5	195
`unlock_prior_online_day`	**0.400**	[0.250, 0.592]	48	0.340	0.352	97%	77.6	195
`random_feasible`	**0.358**	[0.267, 0.458]	43	0.342	0.352	97%	76.4	197
`greedy`	**0.350**	[0.233, 0.467]	42	0.343	0.352	97%	77.3	197
`no_ask`	**0.167**	[0.050, 0.292]	20	0.120	0.352	33%	20.9	0

The four omitted configurations (`fullbudget_unranked_maxw`, identical to `defaultask_core4_maxw` by construction, plus `lab_potential_prior_maxw`, `defaultask_prior_maxw`, `defaultsoft_prior_maxw`) are in `results/primary_summary.csv`, with per-method median and 90th-percentile waiting times.

B.7 Factorial grid and paired contrasts

Cells show MSMI/100 with total events over 60 episodes in brackets; allocation is general-graph maximum-weight matching in every cell.

Ask rule	7-soft agreement count	Signed core-4	Fitted logistic prior
Default / unranked	0.408 (49 ev, n=60)	0.483 (58 ev, n=60)	0.400 (48 ev, n=60)
Count-based potential (incumbent)	0.525 (63 ev, n=60)	0.592 (71 ev, n=60)	0.467 (56 ev, n=60)

Ask rule	7-soft agreement count	Signed core-4	Fitted logistic prior
Unlock-value (ours)	0.525 (63 ev, n=60)	0.492 (59 ev, n=60)	0.400 (48 ev, n=60)

Two allocation ablations sit outside the grid: starter greedy ordering with default asks scores 0.350 (42 events) and with unlock asks 0.475 (57 events).

Contrast	Mean diff	95% CI	p	W/T/L	Events
Global matching vs greedy ordering (soft-7, default asks)	+0.058	[-0.058, +0.167]	0.445	14/40/6	49 vs 42
Signed core-4 vs 7-soft count (default asks)	+0.075	[-0.025, +0.192]	0.328	14/39/7	58 vs 49
Fitted prior vs signed core-4 (default asks)	-0.083	[-0.200, +0.042]	0.266	7/39/14	48 vs 58
Count-based potential vs default asks (soft-7)	+0.117	[-0.017, +0.233]	0.160	22/28/10	63 vs 49
Unlock vs count-based potential (soft-7)	-0.000	[-0.167, +0.200]	1.000	17/26/17	63 vs 63
Unlock vs default (signed core-4)	+0.008	[-0.183, +0.192]	1.000	20/25/15	59 vs 58
Unlock vs default (fitted prior)	-0.000	[-0.142, +0.150]	1.000	15/32/13	48 vs 48
Coding + matching, default asks (vs greedy)	+0.133	[+0.008, +0.258]	0.102	20/33/7	58 vs 42
Coding + matching, unlock asks (vs greedy)	+0.142	[-0.067, +0.358]	0.281	23/23/14	59 vs 42
Pre-specified: incumbent vs official greedy	+0.175	[+0.092, +0.258]	0.016	23/29/8	63 vs 42
Value of clarification (required ablation)	+0.183	[+0.042, +0.325]	0.055	22/33/5	42 vs 20

B.8 Scenario-level table

Method	development	sparse	cold start	delayed	shift	drift	Primary
lab_potential_core4_maxw	0.850	0.150	0.350	0.800	0.650	0.750	**0.592**
lab_potential_maxw	0.800	0.100	0.400	0.550	0.550	0.750	**0.525**
unlock_soft7_maxw	0.700	0.050	0.350	0.850	0.500	0.700	**0.525**
unlock_core4_maxw	0.650	0.100	0.300	0.650	0.650	0.600	**0.492**
defaultask_core4_maxw	0.550	0.050	0.600	0.650	0.450	0.600	**0.483**
unlock_greedy	0.600	0.050	0.450	0.550	0.600	0.600	**0.475**
unlocksoft_prior_maxw	0.450	0.050	0.350	0.700	0.500	0.450	**0.417**
defaultask_soft7_maxw	0.450	0.000	0.600	0.600	0.400	0.400	**0.408**
unlock_prior_maxw	0.500	0.050	0.350	0.550	0.500	0.450	**0.400**
unlock_prior_online_day	0.500	0.050	0.350	0.550	0.500	0.450	**0.400**
random_feasible	0.500	0.050	0.350	0.400	0.350	0.500	**0.358**
greedy	0.350	0.050	0.550	0.400	0.400	0.350	**0.350**
no_ask	0.300	0.000	0.000	0.150	0.250	0.300	**0.167**
coverage ceiling (§3)	0.399	0.177	0.341	0.399	0.399	0.399	

B.9 Funnel counts and stage rates

Method	Assigned	Both responded	Mutual accept	Date	Date ≤30d	Both on time	MSMI
lab_potential_core4_maxw	4624	2753	704	571	567	174	**71**
lab_potential_maxw	4584	2717	727	560	554	156	**63**
unlock_soft7_maxw	4566	2687	676	549	543	145	**63**
unlock_core4_maxw	4629	2819	714	562	560	167	**59**

Method	Assigned	Both responded	Mutual accept	Date	Date ≤30d	Both on time	MSMI
defaultask_core4_maxw	4614	2698	688	534	529	138	**58**
unlock_greedy	4575	2691	681	544	539	133	**57**
unlocksoft_prior_maxw	4666	2747	676	533	532	140	**50**
defaultask_soft7_maxw	4594	2636	669	515	511	122	**49**
unlock_prior_maxw	4652	2869	722	573	572	144	**48**
unlock_prior_online_day	4658	2859	715	577	576	143	**48**
random_feasible	4583	2717	672	535	530	114	**43**
greedy	4641	2710	645	511	508	99	**42**
no_ask	1253	750	180	152	152	47	**20**

Method	P(both respond)	P(mutual given responded)	P(date given mutual)	P(on-time both given date)	P(MSMI given assigned)
lab_potential_core4_maxw	0.595	0.256	0.811	0.305	0.0154
lab_potential_maxw	0.593	0.268	0.770	0.279	0.0137
unlock_core4_maxw	0.609	0.253	0.787	0.297	0.0127
defaultask_core4_maxw	0.585	0.255	0.776	0.258	0.0126
unlock_greedy	0.588	0.253	0.799	0.244	0.0125
unlocksoft_prior_maxw	0.589	0.246	0.788	0.263	0.0107
unlock_prior_maxw	0.617	0.252	0.794	0.251	0.0103
random_feasible	0.593	0.247	0.796	0.213	0.0094
greedy	0.584	0.238	0.792	0.194	0.0090
no_ask	0.599	0.240	0.844	0.309	0.0160

B.10 Cost, waiting times and coverage accounting

Metric	Hard-only ask rules (n=14)	Residual-soft ask rules (n=2)	no_ask
Ask cost per episode (ceiling 720)	196	598	0
Mean first-introduction wait (days)	6.32	6.15	3.95
Median first-introduction wait, of served (days)	3.97	3.64	0.72
90th-pct first-introduction wait, of served (days)	16.02	15.90	13.77
Unserved members of 200	131.6	131.8	176.0
Missing feedback events per episode	39.5	39.7	10.2
Coverage	0.342	0.341	0.120
In-process seconds per episode	6.74	5.83	6.44

Waiting times are computed over **served** members only, as the definition requires; unserved members stay in every coverage denominator. Each cell is the mean over that group's 60 episodes of a per-episode statistic; per-method values are in `results/primary_summary.csv`.

B.11 Residual-budget contrasts and askable-population probe

Contrast	Mean diff	95% CI	p	W/T/L	Events	Ask cost A vs B
Residual soft asks vs hard-only (unlock ranking fixed)	+0.017	[-0.083, +0.158]	0.961	9/41/10	50 vs 48	582 vs 195
Unlock-ranked soft asks vs member-order soft asks	+0.008	[-0.200, +0.208]	1.000	17/28/15	50 vs 49	582 vs 614
Online dense-proxy adaptation vs static prior	+0.000	[+0.000, +0.000]	1.000	0/60/0	48 vs 48	195 vs 195

Members with a complete, never-declined hard bundle (of 200), and live feasible edges between still-free members, averaged over 5 seeds:

Family	Ask rule	d0	d5	d10	d15	d20	d30	d45	d59
development	no_ask · members	50	58	63	68	75	75	75	75
development	default_asks · members	54	82	107	128	140	140	140	140
development	fullbudget_unranked · members	54	82	107	128	140	140	140	140
development	potential_ask_count · members	54	82	107	128	140	140	140	140
development	unlock_value · members	54	82	107	127	139	140	140	140
development	unlock_value_plus_soft · members	54	82	107	127	139	140	140	140
sparse	no_ask · members	49	54	61	67	74	74	74	74
sparse	default_asks · members	53	78	105	121	132	132	132	132
sparse	fullbudget_unranked · members	53	78	105	121	132	132	132	132
sparse	potential_ask_count · members	53	78	105	120	132	132	132	132
sparse	unlock_value · members	53	78	105	119	130	130	130	130
sparse	unlock_value_plus_soft · members	53	78	105	119	130	130	130	130
development	no_ask · live edges	11	1	0	1	0	0	2	0
development	unlock_value · live edges	15	4	4	2	5	1	1	2

All five ranked ask rules are within one member of each other at every day after day 10 in both families; live-edge counts for the other ranked rules are in `results/ask_timing.json`.

FIGURE 07

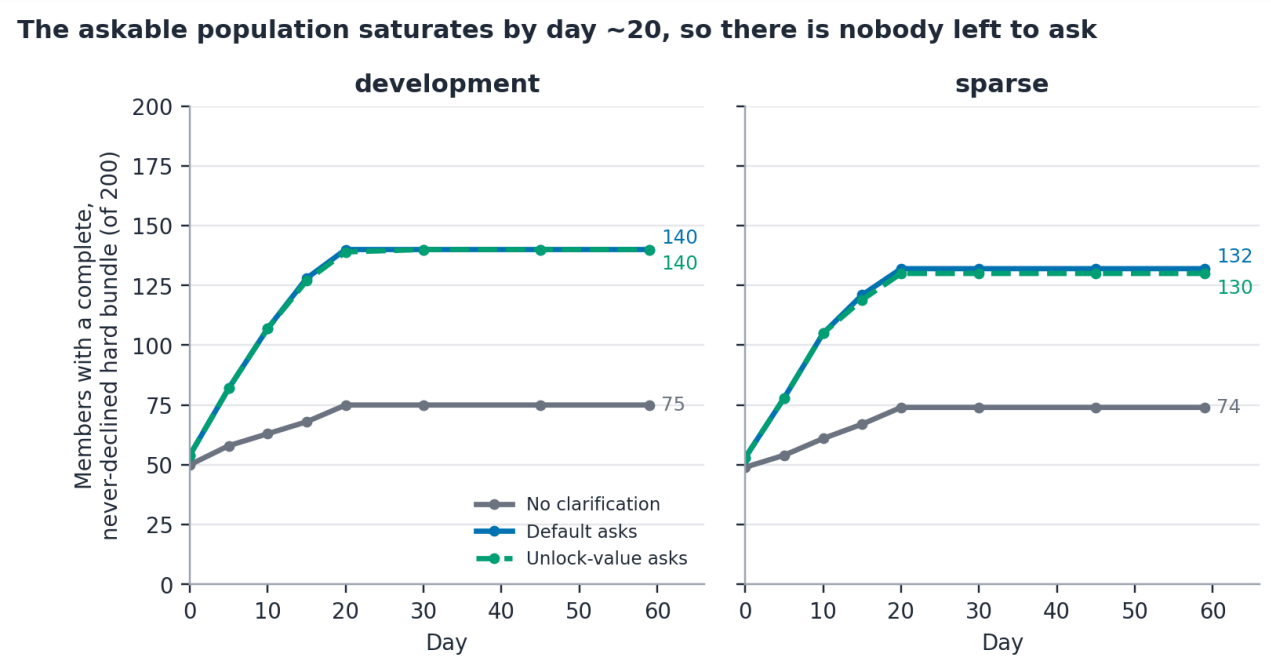


Figure 7. The askable population saturates by day ~20 under every ranked rule (140 of 200 in development, 132 in sparse) while no-clarification plateaus at 75, leaving 40 days of budget with nobody left to ask.

B.12 Oracle ladder per family

180 oracle-ladder episodes over 10 seeds. R4/R5 are analysis-only: they write disclosed answers straight into the simulator's member records, which no policy may do and none of ours does. The latent-pair-probability rung was dropped (§4.6).

Rung	cold_start	development	sparse	Mean	Assignments	Coverage
R0 no clarification + greedy	0.000	0.300	0.000	**0.100**	14.6	0.091
R1 default asks + random feasible	0.350	0.500	0.050	**0.300**	61.9	0.297
R2 official greedy baseline	0.550	0.350	0.050	**0.317**	62.5	0.298
R3 our policy with the fitted prior	0.250	0.450	0.050	**0.250**	63.2	0.297
R4 oracle: free complete clarification	0.100	0.500	0.100	**0.233**	62.9	0.298
R5 oracle: R4 + maximum-cardinality allocation	0.150	0.450	0.100	**0.233**	63.1	0.298

B.13 Attainable primary-score bound

Pair-selection quality	P(mutual given both resp.)	MSMI / assignment	dev	sparse	cold	Primary bound
Randomised-logging quality (measured)	0.2538	0.0105	0.619	0.122	0.475	**0.512**
Incumbent published quality	0.300	0.0124	0.732	0.145	0.562	0.606
Strong selection: the three relevant fields agree	0.400	0.0165	0.976	0.193	0.749	0.808
Best-case selection	0.500	0.0207	1.220	0.241	0.937	1.009

The bound assumes every one of the 118 / 90.6 / 23.3 effective edges is consumed, ignoring arrival and exit timing, the eight-day busy window and retirement on success. The randomised logging policy consumed 71% of the edge supply in development, shift and drift, 68% in delayed, 79% in cold start and 87% in sparse, and scored 0.375 against the 0.512 bound.

B.14 Never-served partition

Family	Never introducible	of which: declined hard field	of which: no feasible partner under complete disclosure	Overlap
development	**60.2%**	31.9%	33.9%	5.6 pp
cold start	**66.0%**	41.0%	32.1%	7.1 pp
sparse	**82.3%**	32.8%	67.0%	17.5 pp

The two columns are measured on different graphs (observed questionnaire vs full disclosure), so they overlap rather than nest; only their union bounds what is achievable.

Appendix C. Baseline equivalence against the reference implementation

Reference mode (`policy.py`)	Our in-process method	Cases	Identical ask lists, pair batches and final metrics
`greedy`	`greedy`	6	**all identical**
`no_asks`	`no_ask`	6	**all identical**
`random`	`random_feasible`	6	**all identical**

Cases cover seeds 101 to 606 across all six families, 60 days each, comparing every ask list and every pair batch day by day.

Appendix D. Validity threats

Threat	Effect	Mitigation
Public seeds only	Private worlds differ; family behaviour may not transfer	Report per family, never only the primary score
Winner's curse	Screening many configurations inflates the best one	Report every configuration screened, including the losers
In-process timing	Not the official 10 s / 2-core / 1 GiB container limit	Report wall clock separately; container test is Round 2 work
Unadjusted multiplicity	p-values are optimistic	State the number of contrasts; give W/T/L as well
Rare events	A zero is not an equivalence result	Report counts and CIs, never only means
Label-cutoff bug (found in our own code)	Online learners may have trained on wrong labels	§7.5